

EEG-to-Image Retrieval under Limited Data: A Reproducible THINGS-EEG Baseline with From-Scratch Generation Stress Tests

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Abstract

Reconstructing visual stimuli from electroencephalography (EEG) is difficult because scalp measurements have high temporal but limited spatial resolution, and paired EEG-image datasets remain small relative to modern generative-model training corpora. We present a reproducible single-subject baseline on THINGS-EEG2 that first tests the more defensible question of whether EEG can retrieve the viewed stimulus in a visual embedding space. A compact temporal-spatial convolutional encoder maps repetition-averaged EEG (63×250) to provided 512-dimensional ViT-B/32 image features. Model selection uses a concept-disjoint validation split, and final evaluation uses the official 200-image, 200-concept test gallery. Across three training seeds, the model obtains $12.83 \pm 0.58\%$, $39.17 \pm 1.76\%$, and $58.00 \pm 1.73\%$ image recall at 1, 5, and 10 (mean $\pm$ sample standard deviation), compared with analytical chance levels of 0.5%, 2.5%, and 5.0%. A session-balanced ablation shows that averaging more test repetitions generally improves ranking. Applying the Subject 01 model to the other nine subjects without adaptation causes a sharp performance drop, exposing subject specificity. We further report exploratory stress tests of direct conditional generators trained without external visual weights. Single-subject and ten-subject variants produce noise-dominated outputs; early decreases in validation ℓ_1 reverse after one to four epochs. Finally, we distinguish direct reconstruction from semantic rendering with a pretrained diffusion prior. The results support reliable coarse semantic decoding under a closed-set, repetition-averaged protocol, but do not support faithful recovery of stimulus pixels. Code is available at <https://github.com/harshit-goyal/brain-to-image>.

1 Introduction

Visual decoding from brain activity asks whether neural measurements recorded during perception retain enough information to identify or reconstruct the viewed content. Electroencephalography is attractive for this purpose because it is non-invasive, portable, inexpensive relative to functional magnetic resonance imaging, and temporally precise. Its low spatial resolution, volume conduction, and sensitivity to artifacts nevertheless make image-level decoding particularly challenging.

Recent work has combined EEG encoders with semantically structured vision representations and powerful pre-trained generators [14, 6, 1]. This strategy can produce visually compelling samples, but the generator contributes substantial prior knowledge. Consequently, a plausible image is not by itself evidence that stimulus-specific shape, texture, pose, or background was recovered from EEG.

This work isolates two questions. First, can a compact EEG encoder recover enough information to rank the correct stimulus above unrelated candidates under concept-disjoint evaluation? Second, can the available paired data train a useful image generator from random initialization? The first question yields a positive retrieval result; the second yields a documented negative result. This separation prevents image realism supplied by a pretrained prior from being mistaken for neural reconstruction fidelity.

Our contributions are:

- a compact and reproducible EEG-to-visual-feature retrieval pipeline for THINGS-EEG2;
- concept-disjoint validation, official 200-concept testing, analytical chance levels, and a shuffled-pair negative control;

- three-seed, session-balanced repetition, and cross-subject robustness analyses; and
- transparent, compute-bounded failure cases for direct single- and multi-subject conditional generation trained without pretrained visual weights.

1.1 Research questions and scope

We organize the study around three research questions:

- RQ1** Does repetition-averaged EEG retrieve an unseen visual concept above closed-gallery chance?
- RQ2** How sensitive is retrieval to the number of EEG repetitions, random initialization, and subject shift?
- RQ3** Under the available image and compute budget, what failure modes arise when the visual generator is trained from random initialization?

RQ1 is the primary confirmatory experiment. RQ2 consists of robustness analyses, while RQ3 is an exploratory stress test. We do not claim open-world, single-trial, cross-subject, or faithful pixel reconstruction.

2 Related Work

EEG visual decoding. Convolutional networks can learn temporal and spatial EEG representations directly from sensor-time arrays [13]. EEGNet [5] introduced a compact design based on temporal, depthwise spatial, and separable convolutions, making it suitable for regimes with limited neural data. Early visual-decoding studies reported classification and generation from EEG representations [15, 4, 9, 8]. Subsequent analysis showed that block-wise acquisition and trial-level splitting can introduce temporal leakage and greatly inflate performance [7]. Leakage arising from non-independent EEG segments has also been documented more broadly [2]. These findings motivate explicit definition of the generalization unit and concept-disjoint evaluation.

Large-scale natural-image EEG. THINGS-EEG2 provides EEG responses to a broad set of natural object concepts under rapid visual presentation [3]. NICE aligns EEG and image representations contrastively and evaluates zero-shot recognition on held-out concepts [14]. Related work uses EEG embeddings and guided diffusion for retrieval and reconstruction [6]. Our study follows this embedding-space view but focuses on a compact, inspectable baseline and explicit controls rather than state-of-the-art comparison.

Visual priors and generation. CLIP learns a semantically organized multimodal representation from large-scale image-text supervision [10]. Latent diffusion models similarly acquire extensive visual knowledge from large image corpora [11]; adversarial diffusion distillation enables few-step synthesis [12]. DreamDiffusion combines masked EEG pretraining, CLIP alignment, and a pretrained diffusion prior [1]. Such systems are valuable for visualization, but their output must be interpreted as a combination of decoded neural evidence and generator prior knowledge.

3 Data and Protocol

3.1 Dataset

We use THINGS-EEG2 [3] through the derived, float16 release distributed with the Uncertainty-Aware Blur Prior (UBP) implementation [16]. The UBP preprocessing epochs raw data from -200 to $1,000$ ms, applies baseline correction through stimulus onset, resamples to 250 Hz, and retains the final 250 samples (0 – $1,000$ ms). Multivariate noise normalization is computed separately by session using only training-partition covariance and is then applied to training and test data. The primary experiment uses Subject 01. Each EEG observation contains 63 channels and 250 temporal samples. The training partition contains $16,540$ unique images from $1,654$ concepts and four EEG repetitions per image. The official test partition contains 200 images from 200 held-out concepts and 80 repetitions per image. The derived release also provides normalized 512 -dimensional CLIP ViT-B/32 image features [10].

For the primary experiment, all available repetitions are averaged before encoding: four during training and 80 during testing. This asymmetry raises test-time signal-to-noise ratio and means the reported results do not measure single-trial decoding. No new human participants were recruited, and this work uses only the released, anonymized dataset under its original data-governance and ethics procedures.

Table 1: Primary Subject 01 data used in this study. Repetitions are averaged into one encoder input per image.

Partition	Images	Concepts	Repetitions/image
Optimization	14,890	1,489	4
Validation	1,650	165	4
Official test	200	200	80

Evaluation pipeline (retrieval; no image synthesis in the primary result)

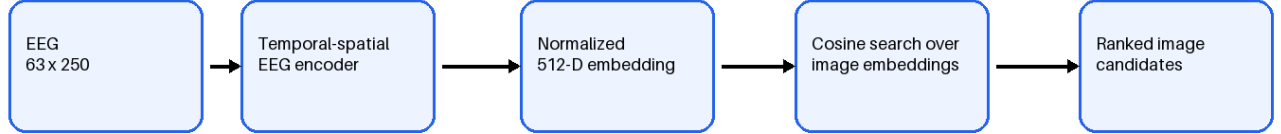


Figure 1: Primary evaluation pipeline. The EEG encoder predicts a normalized visual feature, which is compared by cosine similarity with candidate image features. Image synthesis is not part of the primary retrieval result.

3.2 Leakage-aware split

The official test set remains untouched during training and model selection. Within the training partition, concepts rather than trials or image instances are randomly divided with seed 42: 1,489 concepts (14,890 images) are used for optimization and 165 concepts (1,650 images) for validation. Thus, no validation concept appears in training. This controls semantic memorization in addition to the trial/block leakage concerns identified in prior EEG work [7, 2].

4 Method

4.1 EEG encoder

The encoder follows the compact temporal-spatial pattern of EEGNet [5]. Input $x \in \mathbb{R}^{63 \times 250}$ first passes through a temporal convolution with 32 filters and kernel width 25. A grouped spatial convolution spans all 63 channels and produces 64 feature maps. After batch normalization, GELU activation, average pooling, and dropout, a depthwise temporal convolution of width 15 and a pointwise convolution produce 128 maps. A second pooling stage leaves 15 temporal positions. The flattened representation is projected to 512 dimensions, layer-normalized, and ℓ_2 -normalized.

4.2 Alignment objective

Let $\hat{z}_i$ denote the normalized EEG prediction and z_i the normalized visual feature for item i . For a batch of size B , similarities are

$$s_{ij} = \frac{\hat{z}_i^\top z_j}{\tau},$$

where $\tau = 0.07$. We optimize symmetric cross-entropy over EEG-to-image and image-to-EEG directions, plus cosine alignment:

$$\mathcal{L} = \frac{1}{2} [\text{CE}(S, I) + \text{CE}(S^\top, I)] + 0.25 \left(1 - \frac{1}{B} \sum_i \hat{z}_i^\top z_i \right).$$

Images sharing a concept are masked as off-diagonal negatives so the objective does not explicitly push semantically equivalent images apart.

The model is trained for 12 epochs with AdamW, learning rate 3×10^{-4} , weight decay 10^{-3} , batch size 128, dropout 0.35, and gradient-norm clipping at 1.0. Temporal kernels span 100 ms and 60 ms at the 250 Hz sampling

Table 2: EEG encoder architecture. The model has 999,008 trainable parameters; 98.6% are in the final projection.

Stage	Operation	Output channels	Temporal width
Input	EEG tensor	63	250
Temporal	Conv2D, kernel 1×25	32	250
Spatial	grouped Conv2D, kernel 63×1	64	250
Pool 1	average pool 1×4	64	62
Separable	depthwise 1×15 , pointwise	128	62
Pool 2	average pool 1×4	128	15
Projection	flatten, linear, layer norm	512	1

Table 3: Primary retrieval training configuration.

Setting	Value	Setting	Value
Optimizer	AdamW	Batch size	128
Learning rate	3×10^{-4}	Weight decay	10^{-3}
Epoch budget	12	Dropout	0.35
Temperature	0.07	Cosine weight	0.25
Gradient clip	1.0	Validation fraction	0.10

rate. The fixed 12-epoch budget and remaining hyperparameters were chosen before test evaluation. The checkpoint with the highest validation concept recall at 5 is selected. The final epoch was selected in each of the three seeds and validation had not fully saturated, so the runs should be treated as baselines rather than optimized estimates.

4.3 Retrieval evaluation

At test time, each EEG prediction is compared with all 200 image features by cosine similarity. Recall at k is one when the paired test image appears in the top k candidates. Because the official gallery has one image per concept, image and concept recall are identical on this split. Analytical chance is $k/200$. We additionally apply one deterministic random permutation to EEG-stimulus correspondences and report the resulting shuffled control. Because only one permutation is used, this control is expected to fluctuate around analytical chance and is not a permutation-test estimate.

Metric definitions. For query i , let r_i be the one-indexed rank of its paired image among the gallery. Recall at k is

$$R@k = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[r_i \leq k].$$

We additionally report mean and median rank and paired cosine similarity. Wilson intervals treat the 200 test concepts as Bernoulli observations. The shared candidate gallery introduces dependence, so intervals are descriptive rather than a complete hierarchical uncertainty model.

5 Results

Table 4 and Figure 2 summarize the primary result. The correct image is ranked first for 27 of 200 concepts, within the top five for 82, and within the top ten for 118. The median rank is 8 and the mean rank is 17.11. Mean paired cosine similarity is 0.219.

Under an independent-query binomial approximation, recall at 1 is above chance with $p < 10^{-20}$. The Wilson intervals and this tail probability treat queries as independent, an approximation because all queries share the same gallery. These calculations establish performance above random gallery selection under the stated test protocol; they do not establish cross-subject generalization or rule out every dataset confound.

Table 4: Subject 01 retrieval on the official 200-concept test gallery. Confidence intervals are 95% Wilson intervals over 200 test concepts.

Metric	Model	95% CI	Chance	Shuffled EEG
Recall@1	13.5%	[9.4, 18.9]	0.5%	0.5%
Recall@5	41.0%	[34.4, 47.9]	2.5%	2.0%
Recall@10	59.0%	[52.1, 65.6]	5.0%	3.5%

Official THINGS-EEG Test Retrieval (200 unseen concepts)

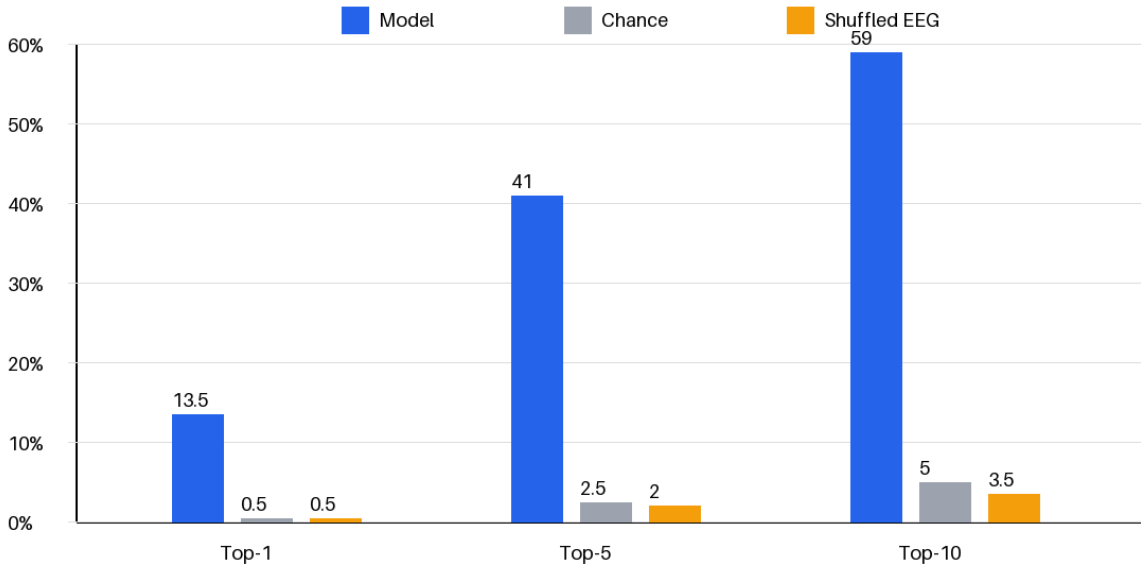


Figure 2: Retrieval accuracy substantially exceeds analytical chance and a fixed shuffled-EEG control at each value of k .

5.1 Training behavior

Concept recall at 5 on the validation partition rises from 4.97% after the first epoch to 21.39% after epoch 12, compared with analytical chance of approximately 3.00%. Validation image recall at 5 reaches 8.12%, compared with image-level chance of approximately 0.30%. The divergence between image and concept retrieval on validation is expected because each validation concept has multiple image instances.

5.2 Robustness analyses

To address RQ2, we repeat the complete Subject 01 training protocol with seeds 7 and 21 alongside the original seed 42. Table 5 shows limited run-to-run variation. Recall@1 ranges from 12.5% to 13.5%, and all runs have a median rank of 8 or 9. The aggregate standard deviations are small relative to the gap from analytical chance, although three seeds remain insufficient for a full characterization of optimization uncertainty.

We next hold the seed-42 weights fixed and vary test-time averaging. The 80 test repetitions comprise four sessions of 20 repetitions. To avoid confounding repetition count with session coverage, each ablation averages an equal number of the earliest repetitions from every session. Consequently, the tested totals are 4, 8, 20, 40, and 80 rather than a prefix of the stored trial axis. As Table 6 and Figure 3 show, mean and median rank improve consistently with additional repetitions. Recall@5 and recall@10 also rise overall, from 24.5% and 37.0% with four repetitions to 41.0% and 59.0% with 80. Recall@1 is non-monotonic, peaking at 15.5% with 20 repetitions; this illustrates that top-1 alone is a noisy summary for a 200-query test set.

Finally, we apply the seed-42 Subject 01 model without fine-tuning to the official test data of Subjects 02–10, averaging all 80 repetitions. This strict transfer test retains the shared channel layout and preprocessing but does

Table 5: Official Subject 01 test results across independently initialized training runs. The last row reports mean $\pm$ sample standard deviation.

Seed	R@1	R@5	R@10	Median rank
42	13.5%	41.0%	59.0%	8
7	12.5%	39.0%	59.0%	8
21	12.5%	37.5%	56.0%	9
Aggregate	$12.83 \pm 0.58\%$	$39.17 \pm 1.76\%$	$58.00 \pm 1.73\%$	8.33 ± 0.58

Table 6: Session-balanced repetition ablation for the fixed seed-42 model. Each condition contributes equally from all four test sessions.

Repetitions	R@1	R@5	R@10	Median rank	Mean rank
4	9.0%	24.5%	37.0%	19	35.87
8	15.0%	31.5%	47.5%	12	26.28
20	15.5%	36.0%	48.0%	11	20.45
40	13.5%	38.0%	56.0%	9	17.76
80	13.5%	41.0%	59.0%	8	17.11

not estimate a subject-adapted mapping. Table 7 and Figure 4 show a large transfer penalty: recall@10 falls from 59.0% on Subject 01 to 15.5–33.5% on the other subjects, while median rank worsens from 8 to 21–48. Every transferred subject remains numerically above analytical chance at recall@10, but these exploratory values should not be read as inferential evidence without subject-level training replications and uncertainty estimates.

6 Generation Stress Tests

We evaluated generation separately from retrieval to avoid conflating visual plausibility with decoded information.

Feature-to-pixel decoder. A transposed-convolution decoder trained to map provided visual features to 64×64 pixels reached a best concept-disjoint validation multi-scale reconstruction loss of 0.1715. When driven by predicted EEG features, outputs preserved broad color and low-frequency layout but were blurred and lacked recognizable object identity. This exposes a distribution mismatch between true visual features used to train the decoder and noisy EEG predictions used at inference.

Single-subject conditional GAN. We jointly trained a temporal–spatial EEG encoder, transposed-convolution generator, and projection discriminator from random initialization, using no external visual weights. The six-epoch run used Adam, batch size 256, generator and discriminator learning rates of 2×10^{-4} and 10^{-4} , and an ℓ_1 weight of 10. Validation ℓ_1 error improved to 0.4442 at epoch 4 and then degraded. A training-set mean-image baseline achieves 0.4382 on the same validation partition, so the generator does not beat this trivial baseline. Across a fixed eight-image test sample, mean pairwise output RMSE is 0.00048, versus 0.3395 among the corresponding target images (all scaled to $[0, 1]$). Generated outputs are therefore nearly invariant and noise-dominated under this training budget.

Ten-subject conditional GAN. To test whether additional neural observations resolve collapse, we trained a subject-conditioned variant across all ten locally available subjects. Subject files were streamed sequentially, with 1,500 images sampled per subject and epoch from the same at-most 14,890-image training pool. Four epochs exposed the model to 60,000 subject–image examples while retaining four averaged repetitions per image. Validation ℓ_1 was best after epoch 1 (0.4480) and subsequently degraded to 0.4565. Mean pairwise output RMSE is 0.0822 on the same fixed eight-image sample, still far below target diversity of 0.3395. Outputs remained noise-dominated. This compute-bounded stress test suggests that adding subjects did not compensate for limited unique visual examples or adversarial instability under our setup. It does not imply that all direct EEG-conditioned generators must fail.

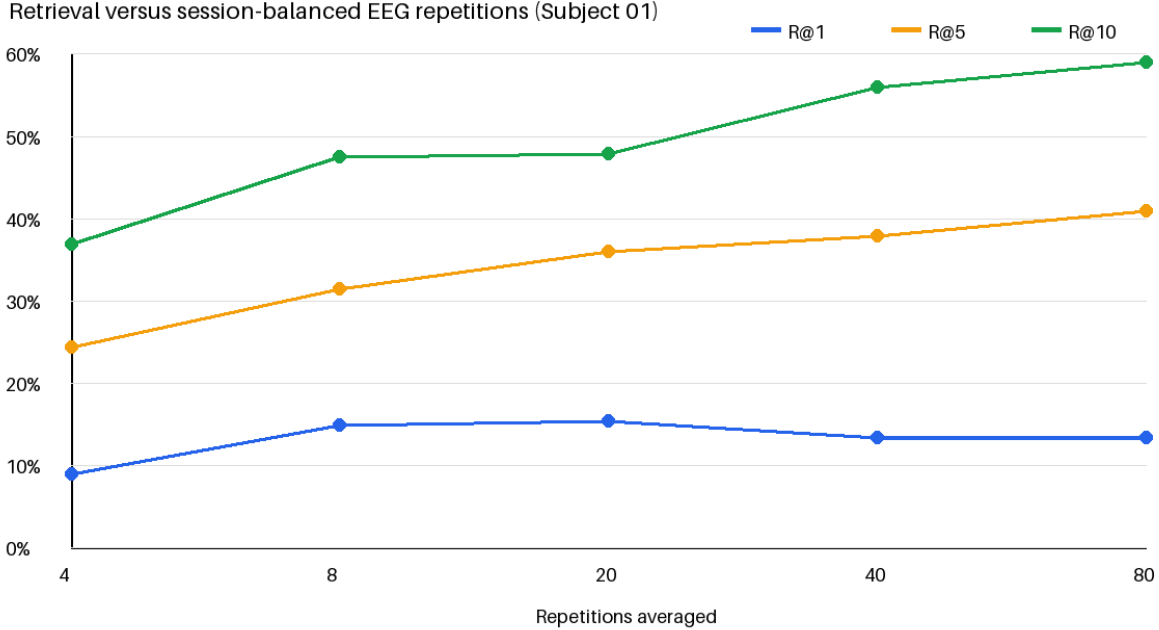


Figure 3: Retrieval versus the number of session-balanced test repetitions. Higher- k recall improves overall as averaging suppresses trial noise.

Table 7: Subject 01 model evaluated without adaptation on all subjects. Analytical chance is 0.5%, 2.5%, and 5.0% at ranks 1, 5, and 10.

Subj.	R@1	R@5	R@10	Med.	Subj.	R@1	R@5	R@10	Med.
1	13.5	41.0	59.0	8	6	4.5	11.5	19.0	39
2	4.0	16.0	24.5	28	7	1.0	10.5	18.5	42
3	2.0	10.5	18.5	48	8	3.0	13.5	20.5	40
4	4.5	14.0	22.5	31	9	2.0	8.5	16.5	46
5	2.5	9.0	15.5	40	10	5.0	20.0	33.5	21

Pretrained diffusion visualization. For qualitative demonstration only, predicted top-1 concepts were converted to text prompts and rendered with SD-Turbo, a distilled pretrained diffusion model [12]. These images are sharp because the diffusion model supplies a large external visual prior. They are semantic renderings of the decoded class, not direct EEG-to-pixel reconstructions, and are excluded from the quantitative claims. The only EEG-derived signal passed to the generator is one discrete class selected from 200 candidates (at most $\log_2 200 \approx 7.6$ bits); no continuous EEG or image feature enters the diffusion model.

7 Discussion

The retrieval result shows that repetition-averaged Subject 01 EEG contains information that generalizes to unseen THINGS-EEG2 concepts in a closed 200-candidate gallery. The gap between recall and chance is large, and the shuffled control is consistent with random correspondence. The experiment therefore supports coarse semantic decoding under this protocol.

The robustness analyses qualify that result. Similar performance across three initializations suggests that the finding is not driven by one fortunate seed. Session-balanced averaging improves rank and higher- k recall, showing that much of the reported accuracy depends on repeated measurements rather than single-trial signal quality. Cross-subject transfer is consistently much weaker than within-subject performance. A deployable decoder would therefore need subject-specific calibration or an explicitly trained subject-invariant representation; neither is demonstrated here.

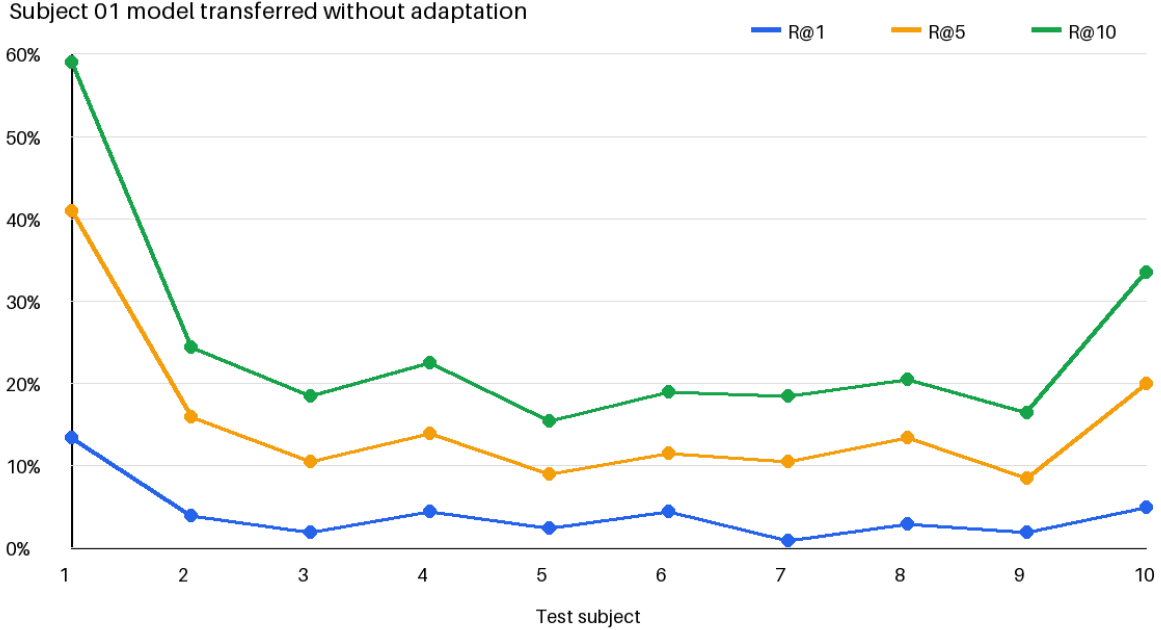


Figure 4: Zero-adaptation transfer of the Subject 01 encoder. Performance drops sharply for every other participant, indicating subject-specific neural-to-visual mappings.

The generation experiments produce a complementary conclusion. Pixel losses encourage average-looking images, while adversarial training with limited paired data is unstable. Increasing the number of subjects adds neural observations but not new image content: the ten-subject experiment draws from the same at-most 14,890-image training pool and only ten images per concept. Learning both a broad natural-image distribution and an EEG-conditioned mapping from this corpus proved unsuccessful.

A stronger future design would train a discrete image autoencoder or generative prior on a substantially larger image corpus, freeze the visual decoder, and then align EEG with its latent representation. If the visual model is pretrained, evaluation should separately quantify what information comes from EEG, for example through retrieval, ablations, shuffled controls, and comparisons against class-only prompts.

8 Limitations

The primary training result is subject-specific and uses only Subject 01. Zero-adaptation transfer to other subjects degrades sharply and is not a substitute for independently trained within-subject models. Eighty test repetitions are averaged for each stimulus, and the repetition ablation starts at four trials to preserve equal session representation; performance should not be interpreted as real-time or single-trial decoding. Evaluation is closed-set: the model ranks a known gallery rather than identifying an arbitrary image from the open world. The study uses provided ViT-B/32 features and therefore inherits their pretrained visual semantics. Three training seeds reduce but do not eliminate optimization uncertainty, and only one shuffled permutation is reported. The repetition conditions reuse the same test concepts and use the earliest trials per session rather than resampling all possible subsets. Hyperparameters were selected for a practical baseline rather than through a comprehensive search. Finally, direct image generation did not succeed; visual realism from the auxiliary diffusion experiment comes primarily from the pretrained generator.

9 Ethical Considerations

Brain-decoding results are vulnerable to sensational interpretation. The present system cannot recover private thoughts, operate without a controlled visual-stimulus protocol, or reconstruct arbitrary images. Its strongest result uses 80 repeated test presentations and a known 200-image gallery. Reporting the work as “mind reading” would

therefore be inaccurate. Potential future applications should require informed consent, clear limits on secondary use, secure handling of neural recordings, and evaluation across demographic and physiological variation. Because this study performs secondary analysis of an anonymized public dataset, it introduces no new participant intervention.

10 Conclusion

A compact EEG encoder can retrieve unseen visual concepts from the official THINGS-EEG2 test gallery substantially above chance when repeated trials are averaged, with similar results across three training seeds. Session-balanced analysis confirms a strong dependence on repeated measurements, while zero-adaptation evaluation reveals substantial cross-subject degradation. In contrast, direct generators trained from scratch on the available paired images collapse or produce blurred, noise-dominated outputs. The key methodological conclusion is that semantic EEG decoding and photorealistic image synthesis must be evaluated separately. Retrieval provides evidence about information present in EEG, whereas a powerful image prior determines much of the appearance of generated samples.

Reproducibility

Training, evaluation, and analysis code given the derived UBP-preprocessed release, along with commands, dependency specifications, tests, configurations, and result artifacts, are available at <https://github.com/harshit-goyal/brain-to-image>. Large dataset files and model checkpoints are excluded from version control. The repository records the random seed, architecture, split procedure, analytical baselines, and fixed shuffled control used for the reported experiment.

A Implementation and Reproduction Details

Experiments were executed with Python 3.9, PyTorch 2.8, and NumPy 2.0 on a MacBook Air with an Apple M2 processor (8 CPU cores), 16 GB unified memory, and the PyTorch MPS backend. Dataset files are loaded as float16 and converted to float32 per sample before repetition averaging. Data-loader workers remain at zero on macOS to avoid copying the approximately 2 GB in-memory subject array into spawned workers.

The main commands are:

```
python3 main.py train --subject 1 --epochs 12 \
  --batch-size 128 --seed 42 --device mps
python3 main.py evaluate --checkpoint outputs/sub-01/best.pt \
  --split test --device mps
python3 paper/analyze_robustness.py --device mps
python3 paper/analyze_seeds.py --device mps
```

Each reported output directory includes a JSON configuration, epoch history, and evaluation metrics. Dataset tensors, downloaded model weights, and checkpoints are omitted from Git because of size; paths and expected tensor shapes are documented in the repository README.

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